VELOOP Rewards - Proposed Analytics Data Pipeline

Proposed design only. Nothing here is claimed to exist in VELOOP today. Solid = start simple; dashed = add only when a scale trigger is met.

Legend

Solid box: Phase 1 (start simple, reuse what exists)

Dashed box: Phase 2 (only when a scale trigger is met)

Red box: quarantine, nothing is deleted

Yellow: supporting components used by several layers

1. Sources and event producers (VELOOP)

Operational database
Users, Tasks, Task_Activity, Rewards, Wallet, Withdrawals, Referrals

Frontend (mobile / web)
interaction events: opened, viewed, started

Third-party providers
ads, surveys, offers, payouts (callbacks)

Backend services
outcome events after commit: completed, credited, reversed

Reference data
approved lists, event catalogue, data dictionary

2. Collection

Scheduled table extract
copies operational tables on a schedule

Event queue / buffer
Phase 2 only: absorbs bursts

Event collection endpoint
envelope, catalogue name, ID and timestamp checks; de-duplicate by event_id

3. Raw zone (append-only, never edited)

raw_table_extracts
dated copies of operational tables

raw_events
exactly what arrived + received_timestamp

Separate warehouse / orchestrator
Phase 2 only

4. Transformation (scheduled SQL, safe to re-run)

Standardise, validate, model
dictionary, approved lists, foreign keys, state transitions (rules DQ001-DQ029)

Reconciliation checks
reward-wallet-withdrawal agreement (DQ030-DQ034)

Quarantine
bad records held with rule ID and reason

5. Analytics layer

Clean entity tables
10 entities in the data dictionary

Quality metrics
completeness, validity, duplicates, orphans

KPI views
one definition per KPI (KPI Catalogue)

6. Reporting

BI / dashboards
7 dashboards, data-as-of banner, quality status

Business reporting
management, product, operations, finance

Monitoring and alerts
job status, freshness, volume, quarantine, reconciliation

read

Phase 2

callbacks

events

events

validate against

dated copies

valid events

rejects

Phase 2

fails

clean

status